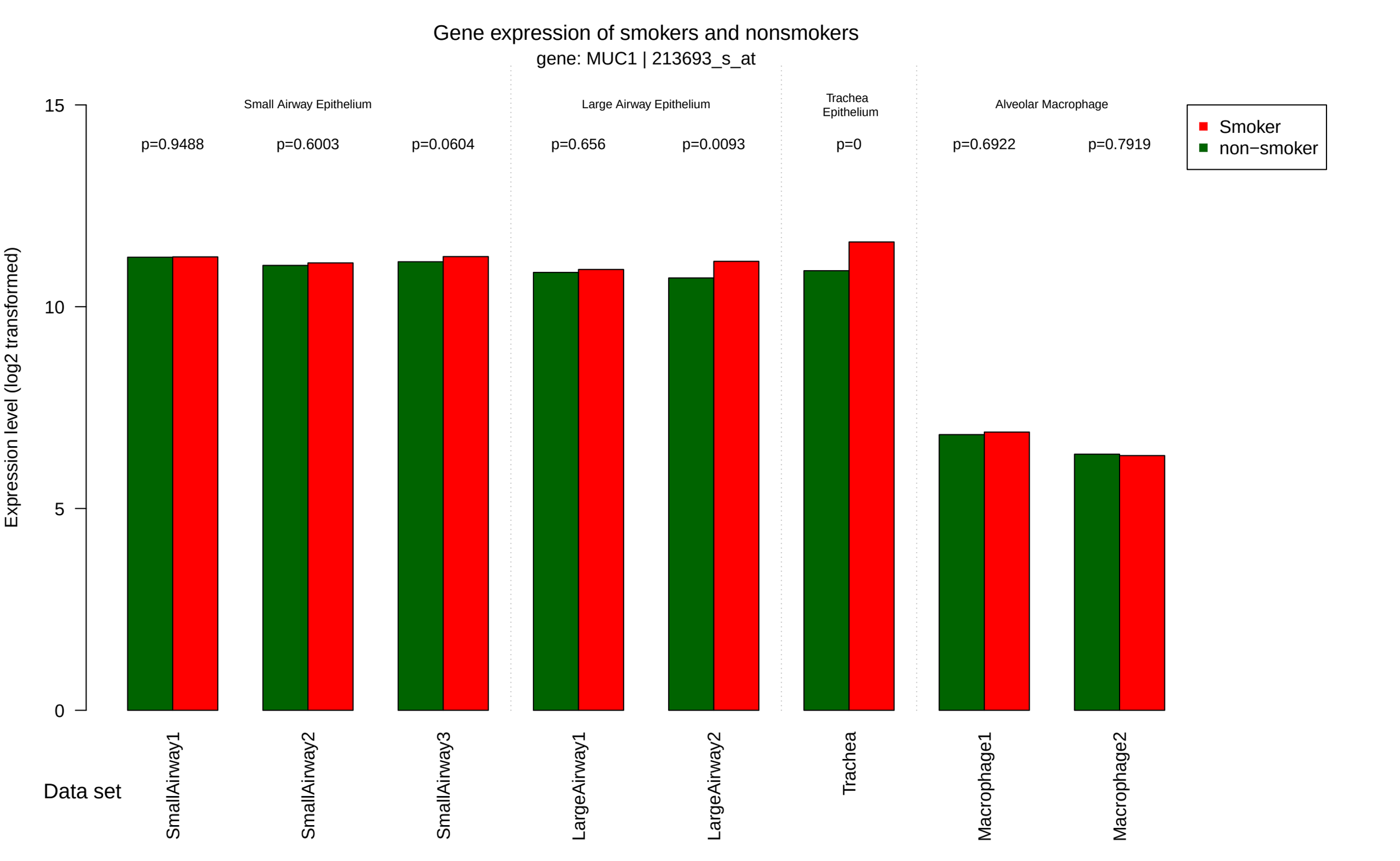
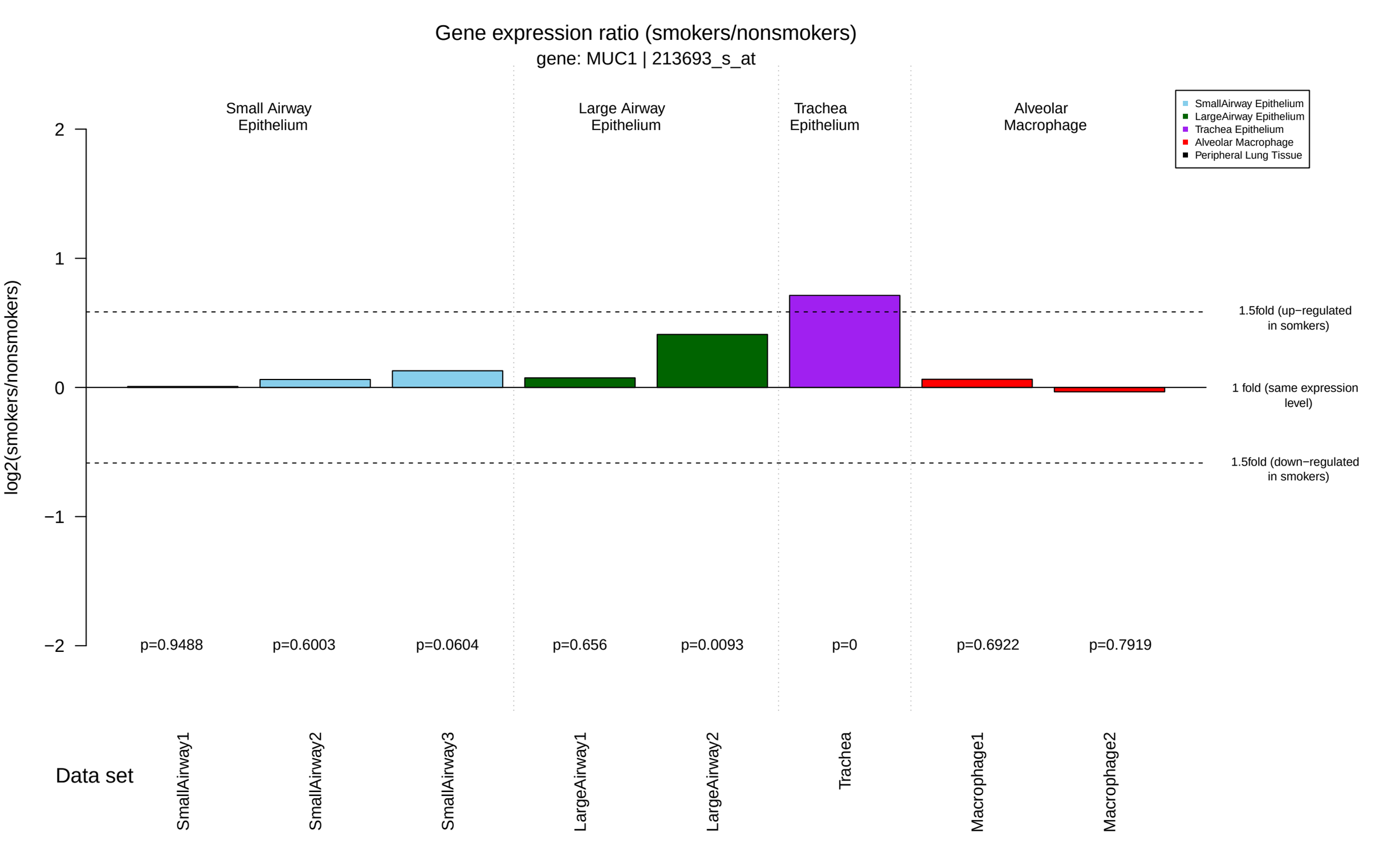
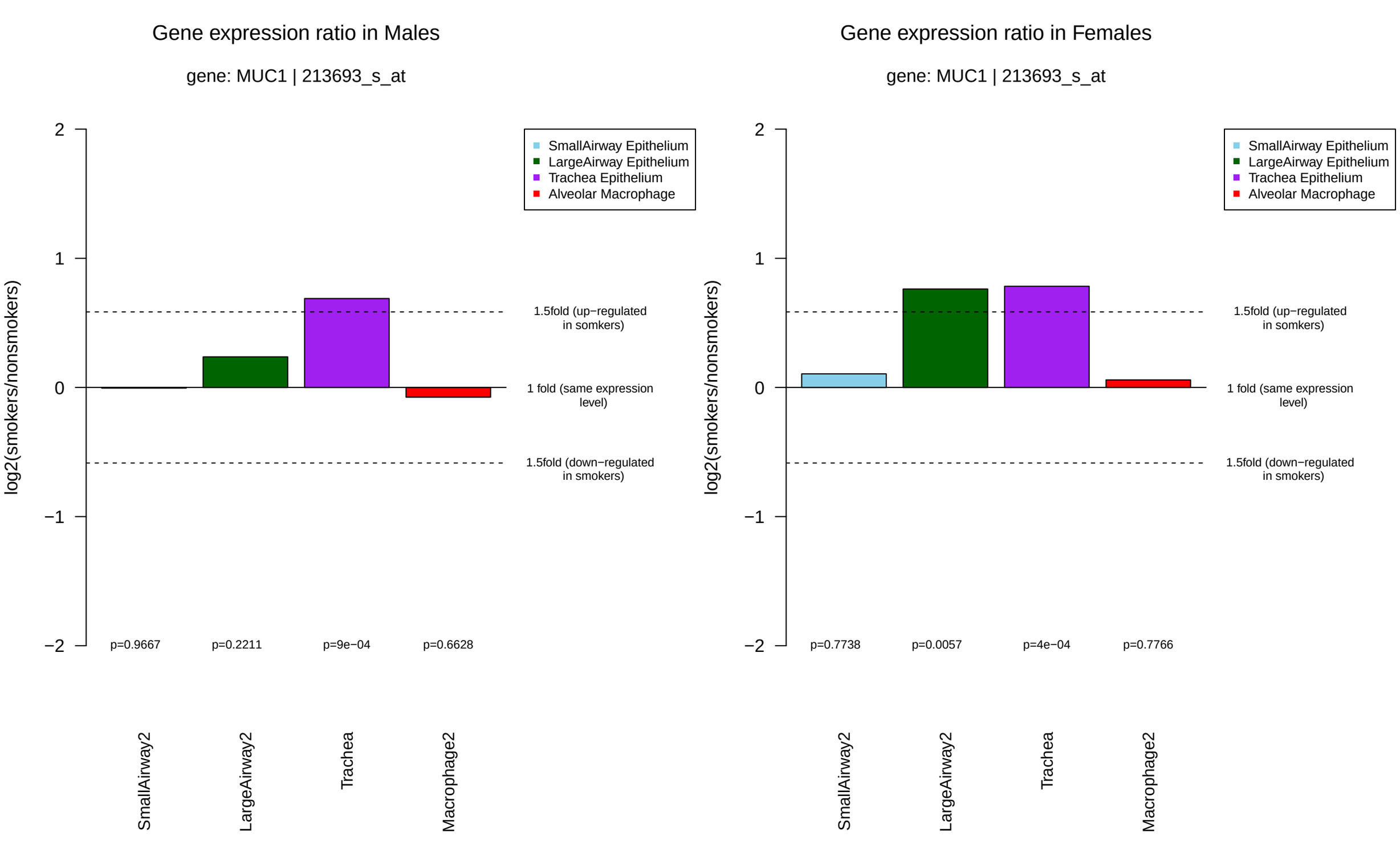
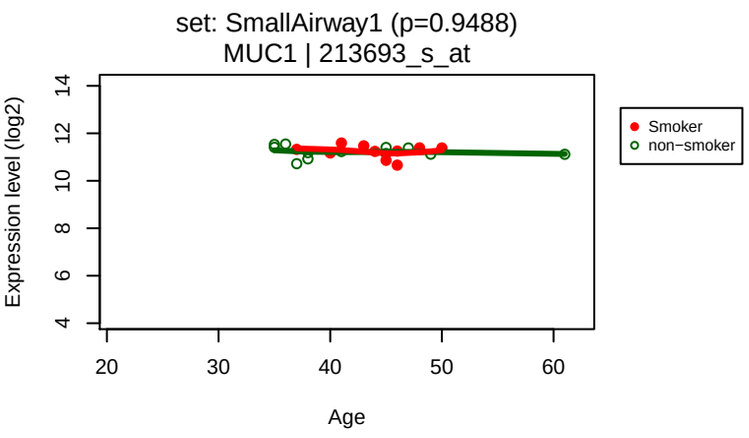
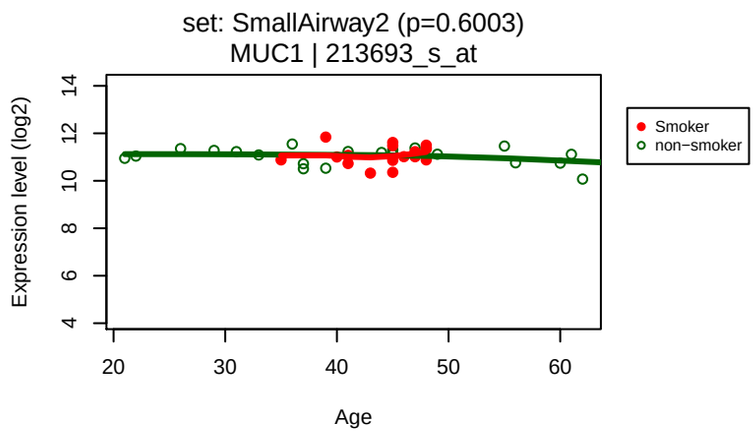
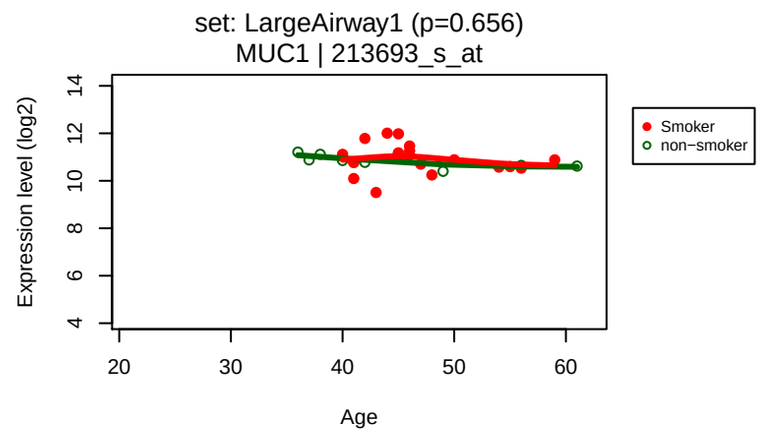

Gene expression of smokers and nonsmokers
gene: MUC1 | 213693_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9488
p=0.6003
p=0.0604
p=0.656
p=0.0093
p=0
p=0.6922
p=0.7919
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

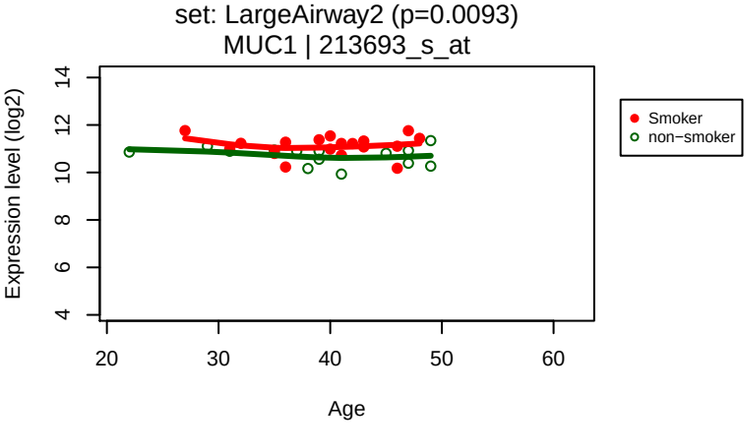

Gene expression ratio (smokers/nonsmokers)
gene: MUC1 | 213693_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9488
p=0.6003
p=0.0604
p=0.656
p=0.0093
p=0
p=0.6922
p=0.7919
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

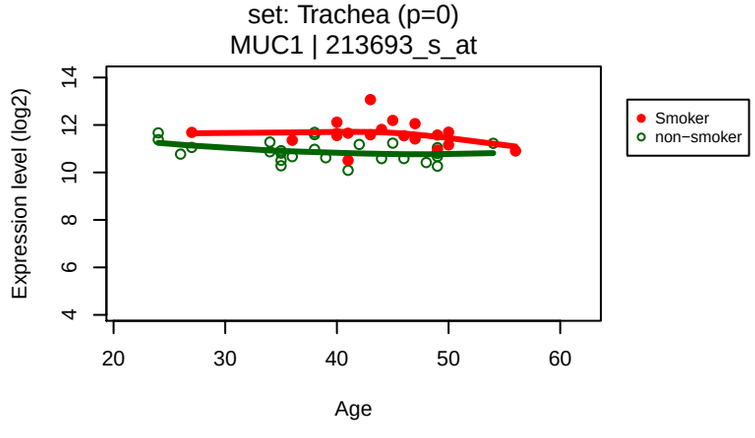

Gene expression ratio in Males
gene: MUC1 | 213693_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9667
p=0.2211
p=9e−04
p=0.6628
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC1 | 213693_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7738
p=0.0057
p=4e−04
p=0.7766
SmallAirway2
LargeAirway2
Trachea
Macrophage2

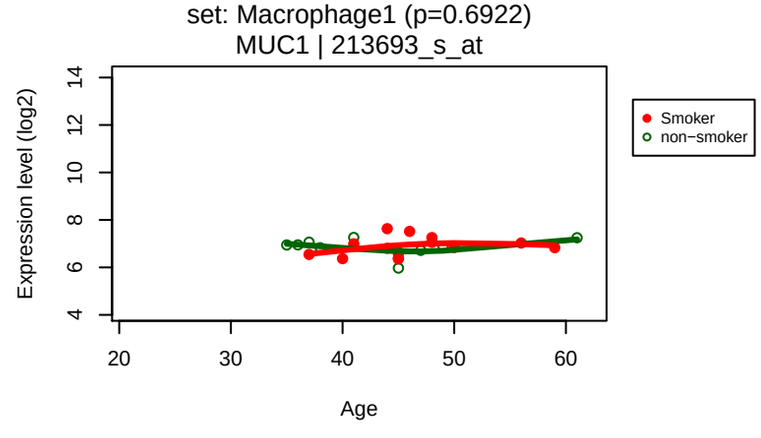

set: SmallAirway1 (p=0.9488)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker

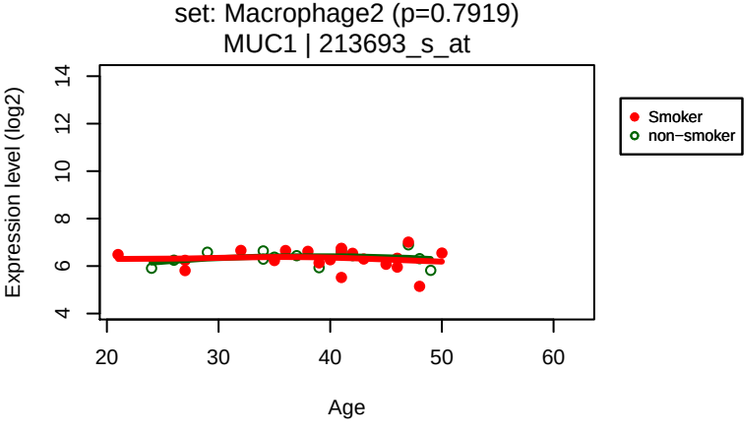

set: SmallAirway2 (p=0.6003)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker

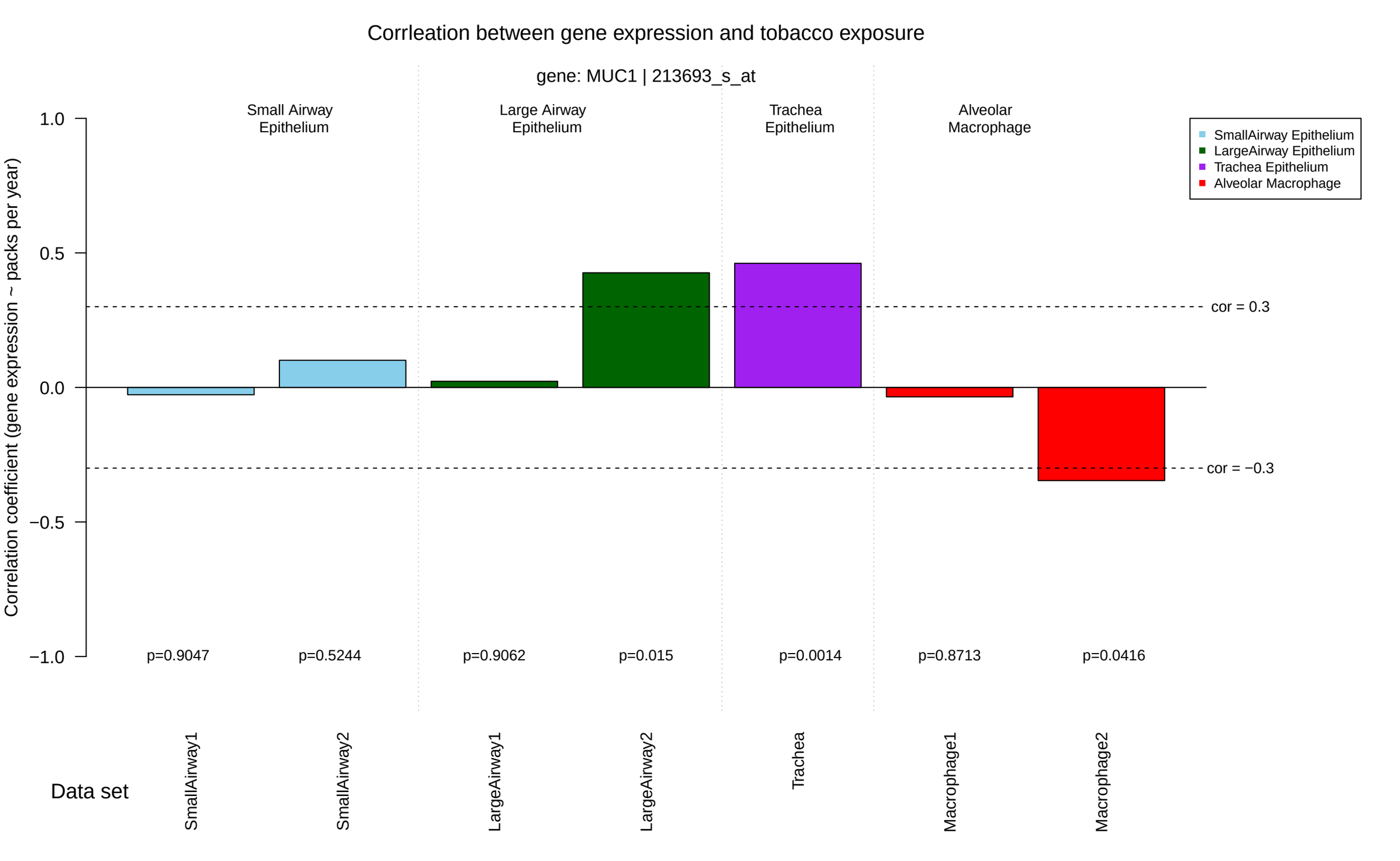

set: LargeAirway1 (p=0.656)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0093)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6922)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7919)
MUC1 | 213693_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC1 | 213693_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9047
p=0.5244
p=0.9062
p=0.015
p=0.0014
p=0.8713
p=0.0416
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2